MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION OCTOBER 2026
FIFTH SEMESTER
DSC- MG5DSCCOM302 - FINANCIAL MANAGEMENT
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Part A

Answer any 12 questions each Question carries 1Marks

1. How is cost of capital used in capital budgeting?

The cost of capital serves as the minimum required rate of return or discount rate for evaluating investment projects. It is used to discount future cash flows in the Net Present Value (NPV) method and to compare a project's return with the required rate of return.

2. What is a debenture?

A debenture is a written instrument acknowledging a company's debt, under which the company agrees to pay interest at an agreed rate and repay the principal according to the specified terms.

3. What is meant by risk-free return?

The risk-free return is the return expected from an investment assumed to have no default risk. The yield on short-term government securities is commonly used as a proxy for the risk-free rate.

4. _____ finance is a branch that studies the emotions and behaviour of investors.

Behavioural finance.

5. A company has an EPS of ₹50 and a retention ratio of 60%. Calculate the dividend payout ratio and dividend per share.

Dividend payout ratio = 100% − Retention ratio = 100% − 60% = 40%.

DPS = EPS × Payout ratio = 50 × 40% = 20 per share.

Payout ratio 40%; DPS 20.

6. What is regular working capital?

Regular working capital is the minimum amount of working capital required continuously to carry on the normal day-to-day operations of a business.

7. _____ is the risk related to variability in a firm's operating income due to business/operating factors.

Business risk (or operating risk).

8. What are mutually exclusive investment decisions?

Mutually exclusive investment decisions arise when a company must choose one among two or more alternative projects, because accepting one excludes the acceptance of the others.

9. Mention any two points indicating the importance of financial leverage.

1. It can increase earnings available to equity shareholders when the return on investment exceeds the after-tax cost of debt.

2. Interest on debt is generally tax-deductible, creating a potential tax advantage.

10. What are the two components of combined leverage?

Operating leverage and financial leverage. Combined leverage measures the combined effect of operating costs and financing costs on earnings per share.

11. What is the significance of a low Degree of Operating Leverage (DOL)?

A low DOL indicates that a given percentage change in sales causes a relatively smaller percentage change in EBIT. It generally indicates lower operating risk and lower sensitivity of operating profit to sales fluctuations.

12. A machine has a book value of 1,20,000 and is sold for 90,000. If the tax rate is 30%, calculate the after-tax salvage value.

Loss on sale = 1,20,000 − 90,000 = 30,000.

Assuming the loss is fully tax-deductible, tax saving = 30,000 × 30% = 9,000.

After-tax salvage value = 90,000 + 9,000 = 99,000.

13. State any two limitations of the payback method.

1. The conventional payback method ignores the time value of money.

2. It ignores cash flows received after the payback period.

14. State any two points highlighting the importance of capital budgeting.

1. It helps a business evaluate and select long-term investment projects.

2. It supports efficient allocation of scarce funds and helps maximise long-term firm value.

Part B

Answer any 4 questions •each question Carries 7 marks

15. Compounded value of an annuity

Annual deposit A = 50,000;

Rate r = 12% = 0.12;

n = 4 years.

Since deposits are made at the end of each year, this is an ordinary annuity.

FV = A × [(1 + r)^n − 1) / r]

FV = 50,000 × [(1.12)^4 − 1) / 0.12]

(1.12)^4 = 1.57351936

FV = 50,000 × 4.779328 = 2,38,966.40

16. Meaning and importance of financial management

Financial management refers to the process of planning, organising, directing and controlling the financial activities of a business. It involves acquiring funds from appropriate sources and using those funds efficiently to achieve the financial objectives of the organisation. It includes investment, financing and dividend decisions.

Importance

- Optimum utilisation of funds
- Procurement of funds
- Investment decisions
- Financing decisions
- Dividend decisions
- Maintenance of liquidity
- Financial planning and control
- Risk management

Financial management is essential for survival, stability and growth. It ensures funds are raised at an appropriate cost and invested in ways that support the firm's objectives.

(**Explanation needed for any 5 points).**

17. Various forms of dividend

A dividend is the portion of a company's distributable profit or reserves distributed to shareholders in accordance with applicable law and the terms of the shares.

- Cash dividend
- Stock dividend / bonus shares
- Property dividend
- Scrip dividend.
- Interim dividend
- Final dividend
- Special dividend
- Liquidating dividend

Dividends may be classified by payment form, timing or purpose. Cash dividends are common, while bonus shares provide additional shares instead of an immediate cash payment

(Explanation needed for any 5 points).

18. Bright Ltd. – Financial leverage after expansion

Existing EBIT = 8, 00,000;

Expected increase = 25%;

Additional debentures = 10, 00,000 at 10%.

Increase in EBIT = 8, 00,000 × 25% = 2, 00,000

EBIT after expansion = 8, 00,000 + 2, 00,000 = 10, 00,000

Interest = 10, 00,000 × 10% = 1, 00,000

EBT = EBIT − Interest = 10, 00,000 − 1, 00,000 = 9, 00,000

Financial leverage = EBIT / EBT = 10, 00,000 / 9, 00,000 = 1.11 times

Financial leverage after expansion is approximately 1.11 times, using the simple EBIT/EBT

19. Profitability Index and comparison with NPV

The Profitability Index (PI), also known as the benefit–cost ratio, measures the present value of future cash inflows per rupee invested.

PI = Present value of cash inflows / Initial investment

Decision rule

PI	Decision
> 1	Accept, subject to other constraints
= 1	Financially indifferent; NPV = 0
< 1	Reject at the specified discount rate

How PI can be useful compared with NPV

- Relative profitability: PI shows present value generated per rupee invested; NPV shows absolute value added.
- Capital rationing: PI can help compare relative attractiveness when investment funds are limited.
- Ranking: PI provides a convenient ratio for ranking independent projects when ratio ranking is appropriate.
- Time value of money: like NPV, PI discounts future cash flows.
- Cash-flow basis: PI uses discounted cash flows rather than accounting profit.

PI is useful for assessing investment efficiency and capital rationing, while NPV remains a fundamental value-maximising criterion.

20. Calculate ARR and select the better project

ARR measures average annual accounting profit as a percentage of investment. As the question provides only initial investment and does not give useful life or residual value, the initial-investment denominator is used.

ARR = (Average annual profit / Initial investment) × 100

Project A

ARR = (₹80,000 / ₹4,00,000) × 100 = **20%**

Project B

ARR = (₹1,25,000 / ₹5,00,000) × 100 = **25%**

Particulars	Project A	Project B
Initial investment	₹4,00,000	₹5,00,000
Average annual profit	₹80,000	₹1,25,000
ARR	20%	25%

Selection: Project B is preferable under the ARR criterion because it provides a higher accounting return of 25%, compared with 20% for Project A.

Mark allocation: Formula (1); Project A calculation (2); Project B calculation (2); comparison and selection (2).

Part C

Answer any 2 questions •each question Carries 15 marks

21. Calculate the Weighted Average Cost of Capital (WACC) **(Value Liberally)**

1.Total capital = ₹10,00,000 + ₹3,00,000 + ₹5,00,000 + ₹2,00,000 = ₹20,00,000

2. Weighted cost of Capital

Source of capital	Amount	Cost %	Weighted Cost %	Weighted Cost of Capital
Equity share capital	1000000	14	7.00	1,40,000
Preference share capital	300000	10	1.50	30,000
Debt	500000	8	2.00	40,000
Retained earnings	200000	12	1.20	24,000
Total	**20,00,000**			**2,34,000**

Weighted Average Cost Of Capital (WACC)= $\frac{\varepsilon WC}{\varepsilon C}=\frac{234000}{200000} * 100$**=11.7%**

OR

Step 1: Total capital

Total capital = ₹10,00,000 + ₹3,00,000 + ₹5,00,000 + ₹2,00,000 = **₹20,00,000**

Step 2: Weights and weighted costs

Source	Capital (₹)	Weight	Cost	Weighted cost
Equity shares	10,00,000	0.50	14%	7.00%
Preference shares	3,00,000	0.15	10%	1.50%
Debt	5,00,000	0.25	8%	2.00%
Retained earnings	2,00,000	0.10	12%	1.20%
Total	20,00,000	1.00		11.70%

Step 3: Weighted Average Cost of Capital

WACC = (0.50 × 14%) + (0.15 × 10%) + (0.25 × 8%) + (0.10 × 12%)

WACC = 7.00% + 1.50% + 2.00% + 1.20% = **11.70%**

The Weighted Average Cost of Capital is 11.70%.

22. What is optimum capital structure? Explain its features.

Capital structure refers to the composition of long-term sources of finance used by a company, generally including equity share capital, preference share capital, retained earnings and long-term debt.

Optimum capital structure is the combination of debt and equity financing that minimizes the overall cost of capital and maximizes the value of the firm, subject to relevant assumptions and risks.

Features

- **Minimum overall cost of capital:** It achieves the lowest feasible weighted average cost of capital for the firm.
- **Maximum firm value:** It supports maximisation of total market value; lower financing costs may increase the present value of future cash flows.
- **Proper balance between debt and equity:** It balances borrowed funds and owners' funds. Excessive debt increases financial risk, while excessive equity may be relatively expensive.
- **Reasonable financial risk:** It avoids an unsustainable debt burden and helps preserve the ability to service obligations.
- **Benefit from financial leverage:** Debt can improve equity earnings when business returns exceed the relevant cost of debt, but it can also magnify losses.
- **Flexibility:** It preserves the ability to raise additional funds for expansion and unexpected needs.

- **Solvency and liquidity:** It maintains the capacity to meet commitments and avoids excessive default risk.
- **Control considerations:** Equity can dilute voting control; debt usually has no equivalent voting rights, though lenders may impose covenants.
- **Tax advantages of debt:** Interest is generally deductible for corporate tax purposes subject to applicable rules, which may reduce the effective cost of debt.
- **Business risk:** Firms with unstable earnings generally have less capacity to support high fixed interest commitments.
- **Creditworthiness and market conditions:** Interest rates, investor expectations, credit ratings and financial position affect the cost and availability of finance.
- **Compatibility with long-term objectives:** The structure should support growth, investment, dividend policy and overall strategy.

Optimum capital structure is not simply the structure with the highest debt or lowest equity. It is the financing mix that balances cost, risk, control, flexibility and solvency while supporting firm-value maximization.

23. Calculate the Net Present Value (NPV) of the project (Value Liberally)

*It is an unconventional project as I involve multiple cash inflows.
*Salvage value should be added to the cash inflow of the final year

Calculation of present value of cash outflows

Year	Cash Outflow	PV Factor	PV of Cash Outflow @10%
0	300000	1	300000
1	60000	0.909	54540
Present value of cash outflow			**354540**

Calculation of present value of cash Inflows

Year	Cash In flow	PV Factor	PV of Cash Inflow @10%
1	40000	0.909	36630
2	60000	0.826	49560
3	120000	0.751	90120
4	160000	0.683	109280
5	60000	0.621	37260
5(Salvage)	80000	0.621	49680
Present value of cash Inflow			**372260**

NPV = Present value of cash inflows − Present value of cash outflows

NPV = ₹3,72,260 − ₹3,54540 = **₹17,720**